

Inverse Mechanical Counting and Mathematics

Part V

Standard Theorems and Relations Shown Operating Inside the Single Count
Worked Examples on the Master Count of the $\sqrt{2}$ Master Mechanic

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Abstract

This paper carries forward the framework established in the first paper of this series, *Inverse Mechanical Counting and Mathematics*. The axiom of 1, the $\sqrt{2}$ master mechanic, the master table, the count-by-count unfolding, the four key numbers (512, ≈ 514.90 , 720, ≈ 724.077), and the four inward vectors read at four operational depths are taken as given. The purpose of this paper is to show standard theorems and relations of mathematics operating inside the master count — not derived from outside it, not imposed on it, but read directly from the count where each relation sits. The Pythagorean theorem is the first worked example: shown to sit at count 3 of the master table, where the right angle is machined from the 720 angular cycle at 90, the legs are the rational reading at even depth $d = 4$, and the hypotenuse is the irrational reading at odd depth $d = 5$ with its $\sqrt{2}$ factor forced by parity. The relation $a^2 + b^2 = c^2$ falls out of the parity rule directly: even-depth squared stays rational, odd-depth squared produces the $\times 2$ that balances the equation. Rational scaling carries the relation to all other right triangles, including standard external triples such as 3, 4, 5. Further classical theorems and relations are worked in the same way as the paper continues, each shown as the mechanical signature of a specific count, depth, and parity within the single first count of the master mechanic.

Scope of this paper. This paper carries the framework of paper one forward and applies it. No new mechanics are introduced. The axiom, the master mechanic, the master table, and the four inward vectors are taken as established. Each worked example takes one standard theorem or relation of mathematics and shows how it operates inside the master count — which count it sits at, which depths its components occupy, and how its standard form is forced by the parity rule and the angular machined cycle. Physical applications belong to Inverse Spherical Dual Hemisphere Quantum Operational Mechanical Theory (ISDHQOM Theory) and remain reserved for future work.

The Axiom of 1 and Its Master Mechanic (Carried Forward)

The Axiom. There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced separately.

The Master Mechanic. The $\sqrt{2}$ mechanic is the master mechanic of the axiom — the simplest mechanic, the minimum partition beyond unity. It reads the axiom inward by halving, under the parity rule:

- Even depth d : the reading is rational.
- Odd depth d : the reading carries a natural factor of $\sqrt{2}$.

Operational Reading. Every reading in this paper carries three coordinates of the master mechanic acting on the axiom: the mechanic unit ($\sqrt{2}$), the operational depth d , and the count position c . The notation $\sqrt{2^d}$ at count c records all three. Standard mathematical relations shown in this paper are read by locating the coordinates each relation occupies on the master count and reading the parity rule at those coordinates.

The mathematics is read, not calculated.

Standard theorems are the readings the master count forces.

1 The Master Count — Carried Forward

The master table is the single source. Every worked example in this paper reads from this table; nothing is added from outside. The **Mechanic** column shows the operational form $\sqrt{2^d}$ at each row's depth: count 0 owns no active mechanic (axiom at rest at $\sqrt{2^0}$), count 1 owns $\sqrt{2^1}$, count 2 owns $\sqrt{2^2}$ at its even depth and $\sqrt{2^3}$ at its odd depth, count 3 owns $\sqrt{2^4}$ and $\sqrt{2^5}$, count 4 owns $\sqrt{2^6}$ and $\sqrt{2^7}$, and count 5 owns $\sqrt{2^8}$ and $\sqrt{2^9}$.

The four key numbers carried forward. The rational closure 512 at $c = 5$, $d = 8$; the sphere boundary ≈ 724.077 at $c = 5$, $d = 9$; the machined angular seat $720 = 5 \times 9 \times 16$ between them; and the derived symmetric counterpart ≈ 514.90 on the square side.

The angular machined cycle carried forward. On the 720 angular vector, the count-by-count halvings produce the natural angular landings: 720 at $c = 0$, 360 at $c = 1$, 180 at $c = 2$, 90 at $c = 3$, 45 at $c = 4$, 22.5 at $c = 5$.

This is everything from paper one that the worked examples in this paper depend on. The first worked example follows.

2 Worked Example 1 — The Pythagorean Theorem on the Master Count

The Pythagorean theorem is the first standard theorem worked. It is not imposed as the rule $a^2 + b^2 = c^2$. It is the mechanical signature of squaring a dual count under the parity rule, forced specifically at count 3 where the angular cycle produces the right angle.

c	Mechanic	d	Parity $\rightarrow$ Reading	Reading value (decimal)
0	no mechanic (axiom)	0	even $\rightarrow$ rational	1
1	$\sqrt{2^1}$	1	odd $\rightarrow$ irrational	$2\sqrt{2} \approx 2.828427$
2	$\sqrt{2^2}$	2	even $\rightarrow$ rational	8
2	$\sqrt{2^3}$	3	odd $\rightarrow$ irrational	$8\sqrt{2} \approx 11.313708$
3	$\sqrt{2^4}$	4	even $\rightarrow$ rational	32
3	$\sqrt{2^5}$	5	odd $\rightarrow$ irrational	$32\sqrt{2} \approx 45.254834$
4	$\sqrt{2^6}$	6	even $\rightarrow$ rational	128
4	$\sqrt{2^7}$	7	odd $\rightarrow$ irrational	$128\sqrt{2} \approx 181.019336$
5	$\sqrt{2^8}$	8	even $\rightarrow$ rational	512
5	$\sqrt{2^9}$	9	odd $\rightarrow$ irrational	$512\sqrt{2} \approx 724.077$

Table 1: The master table with the explicit Mechanic column, carried forward from paper one. Each row shows the operational form $\sqrt{2^d}$ at that depth. Count 0 owns no active mechanic (axiom at rest, $\sqrt{2^0} = 1$). Count 1 owns the singular $\sqrt{2^1}$. Counts 2, 3, 4, 5 each own two consecutive depths — one even (rational) and one odd (irrational) — with the mechanic at each depth shown on its own row. Five active counts unfold nine active depths. The reading value is the operational total at each row, computed as $2^c \times \sqrt{2^d}$.

The Theorem Walked Count by Count

At every count from $c = 1$ onward, the master mechanic produces a dual reading: one rational at the even depth, one irrational at the odd depth. The rational reading squared stays rational; the irrational reading squared produces the $\times 2$ that doubles the rational. When the two equal rational readings are taken as legs and the irrational reading is taken as the hypotenuse, the squared sum equals the squared hypotenuse exactly — this is the Pythagorean relation.

The right angle itself is forced separately by the angular machined cycle. The angular cycle reaches 90 at count 3 ($720/8$), and only at count 3. Counts 1, 2, 4, 5 produce the Pythagorean relation through their dual readings, but they do not produce the right angle. Count 3 is where both pieces come together: the right angle on the 720 angular vector and the leg/hypotenuse dual on the $\sqrt{2}$ master mechanic.

The table below walks every count and shows what the master mechanic produces at each.

The Right Angle Seat — Forced at Count 3

On the 720 angular machined vector, the count-by-count halvings carried forward from paper one produce the natural angular landings at each count. At count 3:

$$\text{Angle at } c = 3 : \quad \frac{720}{8} = 90.$$

Count 3 partitions the angular cycle into eight pieces of 90 each (the octant: $8 \times 90 = 720$). This is the mechanical seat of every right angle. No external 90 is introduced; the right angle is the natural angular landing of the master mechanic at count 3.

The Two Legs — Rational Reading at Count 3 (Even Depth)

Count 3 owns local depths $d = 4$ (even) and $d = 5$ (odd). The even depth gives the rational reading:

$$c = 3, \quad d = 4 : \quad \sqrt{2^4} = 4 \quad (\text{rational}).$$

This is the mechanical seat of the two equal legs when read squarely. Both legs sit on the even rational reading at count 3.

Count	Master Table Reading	Triangle Sides	Right Angle	Squaring Check	Status
$c = 0, d = 0$	$\sqrt{2^0} = 1$ (axiom, rational)	Legs of the $1, 1, \sqrt{2}$ triangle	— (axiom at rest)	$1^2 + 1^2 = 2$	Forced — both legs on the axiom seat
$c = 1, d = 1$	$\sqrt{2^1} = \sqrt{2}$ (singular halving, irrational)	Hypotenuse of the $1, 1, \sqrt{2}$ triangle	$720/2 = 360$ — not yet right angle	$(\sqrt{2})^2 = 2$	Forced — hypotenuse at the first partition
$c = 2, d = 2, 3$	$\sqrt{2^2} = 2, \sqrt{2^3} = 2\sqrt{2}$	Legs $2, 2$ and hypotenuse $2\sqrt{2}$	$720/4 = 180$ — not yet right angle	$2^2 + 2^2 = 8 = (2\sqrt{2})^2$	Forced — first dual count produces an isosceles right triangle reading
$c = 3, d = 4, 5$	$\sqrt{2^4} = 4, \sqrt{2^5} = 4\sqrt{2}$	Legs $4, 4$ and hypotenuse $4\sqrt{2}$	720/8 = 90 — right angle forced	$4^2 + 4^2 = 32 = (4\sqrt{2})^2$	Forced — the canonical right-angle seat of the master count
$c = 4, d = 6, 7$	$\sqrt{2^6} = 8, \sqrt{2^7} = 8\sqrt{2}$	Legs $8, 8$ and hypotenuse $8\sqrt{2}$	$720/16 = 45$ — past the right angle	$8^2 + 8^2 = 128 = (8\sqrt{2})^2$	Forced — self-similar repeat at finer count
$c = 5, d = 8, 9$	$\sqrt{2^8} = 16, \sqrt{2^9} = 16\sqrt{2}$	Legs $16, 16$ and hypotenuse $16\sqrt{2}$	$720/32 = 22.5$ — closure	$16^2 + 16^2 = 512 = (16\sqrt{2})^2$	Forced — closure of the chain

Table 2: The Pythagorean theorem walked count by count on the master table. At every count with a dual reading, the Pythagorean *relation* is forced by the parity rule: even-depth squared stays rational, odd-depth squared produces $\times 2$, and twice the rational leg squared equals the hypotenuse squared. The *right angle* is forced specifically at count 3, where the 720 angular machined cycle lands at 90. Count 3 is the count where both pieces of the theorem come together.

The Hypotenuse — Irrational Reading at the Same Count (Odd Depth)

At the same count 3, the odd depth gives the irrational reading:

$$c = 3, \quad d = 5 : \quad \sqrt{2^5} = 4\sqrt{2} \quad (\text{irrational, } \sqrt{2} \text{ factor}).$$

This is the mechanical seat of the hypotenuse. It carries the $\sqrt{2}$ factor because it sits on the odd depth, which the parity rule forces to carry $\sqrt{2}$.

The Relation $a^2 + b^2 = c^2$ — Forced by Parity

Take the count-3 readings:

- Leg a : 4 (rational, even depth)
- Leg b : 4 (rational, even depth)
- Hypotenuse c : $4\sqrt{2}$ (irrational, odd depth)

Square each reading:

- $a^2 = 4^2 = 16$ (rational squared, stays rational)
- $b^2 = 4^2 = 16$ (rational squared, stays rational)
- $c^2 = (4\sqrt{2})^2 = 16 \times 2 = 32$ (the $\sqrt{2}$ factor squared gives $\times 2$, doubling the rational)

The relation falls out directly:

$$16 + 16 = 32.$$

This is forced by the parity rule. Even-depth (rational) readings squared remain rational. Odd-depth ($\sqrt{2}$ -carrying) readings squared produce the $\times 2$ factor that doubles the rational, balancing the sum of two equal rational squares. The theorem is the mechanical signature of squaring a dual count under parity.

External Examples — The 3, 4, 5 Triangle

The canonical 3-4-5 triangle has its right angle at count 3 (every right triangle uses the same 90 seat). The standard squaring check holds:

$$3^2 + 4^2 = 9 + 16 = 25 = 5^2.$$

However, the integers 3 and 5 do not sit on the master count. They are not powers of 2, not $\sqrt{2}$ times a power of 2, and not on any row of the master table. They are external integers that satisfy the Pythagorean relation that the master count has already forced through the 4, 4, $4\sqrt{2}$ reading at count 3.

Rational scaling preserves the squaring relation, which is why infinitely many integer triples satisfy $a^2 + b^2 = c^2$. The 3, 4, 5 triple is a famous instance of this scaling. The mechanical content of the theorem is in the relation, not in the specific external integers. The forced readings of the master count are the cases where all three sides sit on the master table directly — 1, 1, $\sqrt{2}$ at counts 0 and 1; 4, 4, $4\sqrt{2}$ at count 3 (the right-angle seat); and the self-similar repeats at counts 2, 4, 5.

Why the Theorem Is Forced, Not Assumed

The Pythagorean theorem is not a rule imposed on the count from outside. It is the mechanical signature of reading a right triangle at count 3 of the master count:

- The right angle is the angular landing at $c = 3$ on the 720 vector ($720/8 = 90$).
- The two legs are the rational readings at the even depth $d = 4$ that count 3 owns ($\sqrt{2^4} = 4$).
- The hypotenuse is the irrational reading at the odd depth $d = 5$ that count 3 also owns ($\sqrt{2^5} = 4\sqrt{2}$).
- Squaring an odd-depth $\sqrt{2}$ -carrying reading produces the $\times 2$ that balances the equation.
- Rational scaling preserves the relation, carrying it to all other right triangles.

The theorem operates inside the single count of 5. It is the master mechanic reading itself at count 3, with the parity rule doing all the work.

Summary — The Pythagorean Reading

The right angle lives at count 3, where the 720 angular cycle lands at 90. The legs live on the rational (even) side at count 3. The hypotenuse lives on the irrational (odd) side at count 3. The $\sqrt{2}$ mechanic and the parity rule produce $a^2 + b^2 = c^2$ automatically through the squaring of the dual. Rational scaling carries the relation to all right triangles, including external integer triples such as 3, 4, 5. The theorem is the count, read.

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